

CBSE Class 10 Mathematics (Standard) Board Exam 2018

Question Paper (Questions 1–31)

SECTION A – Multiple Choice Questions (1 Mark Each)

Q1. The HCF and the LCM of 14, 21 and 77 respectively are:

- (A) 7, 77
- (B) 14, 462
- (C) 7, 462
- (D) 21, 77

Q2. The value of x , for which the polynomials $(9 - x^2)$ and $(6x + x^2 + 9)$ vanish simultaneously, is:

- (A) 3
- (B) 2
- (C) -2
- (D) -3

Q3. If the square of the difference of the zeroes of the quadratic polynomial $(y^2 + py + 36)$ is 81, then the values of p are:

- (A) ± 5
- (B) ± 15
- (C) ± 18
- (D) ± 12

Q4. If the first term of an A.P. is -12 and the common difference is 4, then the sum of its first 7 terms is:

- (A) -24
- (B) -48
- (C) 0
- (D) 48

Q5. If the system of equations

$$3x + 2y = 4$$

$$4ax + (a + b)y = 16$$

has infinitely many solutions, then:

- (A) $5a = 3b$
- (B) $3a = 5b$
- (C) $a + b = 15$
- (D) $a - b = 2$

Q6. The distance between the points $((4\cos\theta + 3\sin\theta, 0))$ and $((0, 4\sin\theta - 3\cos\theta))$ is:

- (A) 25
- (B) 7
- (C) 5
- (D) $(\sqrt{7})$

Q7. The point on the y-axis equidistant from A(1, 3) and B(4, 4) is:

- (A) (0, 11)
- (B) (11, 0)
- (C) (0, 13)
- (D) (0, 12)

Q8. If a regular hexagon is inscribed in a circle of radius 3 cm, then its perimeter is:

- (A) 9 cm
- (B) 18 cm
- (C) 27 cm
- (D) 36 cm

Q9. Two circles of radii 10 cm and 17 cm intersect at P and Q. If $PQ = 16$ cm, then the distance AB is:

- (A) 30 cm
- (B) 12 cm
- (C) 21 cm
- (D) 16 cm

Q10. If $\triangle ABC \sim \triangle PQR$ such that $AB = 6$ cm, $AC = 7$ cm, $QR = 15$ cm and $PQ = 12$ cm, then $BC + PR$ equals:

- (A) 44 cm
- (B) 21.5 cm
- (C) 21 cm
- (D) 29.5 cm

Q11. If $(\sin\theta - \cos\theta = 0)$, then the value of $(\sin^6\theta + \cos^6\theta)$ is:

- (A) 1
- (B) $\frac{1}{8}$
- (C) $\frac{3}{4}$
- (D) $\frac{1}{4}$

Q12. If $(\sin\theta + \cos\theta = \sqrt{2} \cos\theta)$, then $(\tan\theta)$ is:

- (A) $(\sqrt{2} + 1)$
- (B) $(\sqrt{2} - 1)$
- (C) $(-\sqrt{2})$
- (D) $(\sqrt{2})$

Q13. A tree 8 m high casts a shadow 6 m long. At the same time a flag pole casts a shadow 30 m long. The height of the flag pole is:

- (A) 40 m
- (B) 22.5 m
- (C) 44 m
- (D) 22 m

Q14. In a circle of radius 14 cm, the area of the sector made by an arc of length 11 cm is:

- (A) 154 cm^2
- (B) 102.67 cm^2
- (C) 205.33 cm^2
- (D) 77 cm^2

Q15. A sector is cut from a circular sheet of radius 50 cm, the central angle being 90° . If another circle of the same area as the sector is formed, then the radius of the new circle is:

- (A) 25 cm
- (B) 50 cm

- (C) 12.5 cm
- (D) 20 cm

Q16. The median of a set of 15 distinct observations is 30.5. If each of the largest 7 observations is increased by 3, then the median of the new set:

- (A) Is increased by 3
- (B) Is decreased by 3
- (C) Is three times the original median
- (D) Remains the same

Q17. The probability that a 2-digit number less than 20 selected at random is a multiple of 2 and not a multiple of 3 is:

- (A) $\frac{1}{2}$
- (B) $\frac{1}{5}$
- (C) $\frac{3}{10}$
- (D) $\frac{3}{11}$

Q18. A card is drawn at random from a deck of 52 playing cards. The probability that the drawn card is not a red face card is:

- (A) $\frac{3}{26}$
- (B) $\frac{23}{26}$
- (C) $\frac{7}{52}$
- (D) $\frac{23}{52}$

Assertion–Reason Questions

Q19.

Assertion (A): The point $(-2, 4)$ divides the line segment joining $(-4, 8)$ and $(5, -10)$ in the ratio $2 : 7$ internally.

Reason (R): If three points P, Q and R are collinear, then $PQ + QR = PR$.

Q20.

Assertion (A): If the total surface area of a solid hemisphere is 462 cm^2 , then its radius is 7 cm.

Reason (R): The total surface area of a solid hemisphere of radius r is $(3\pi r^2)$.

SECTION B – Very Short Answer Questions (2 Marks Each)

Q21. Find HCF and LCM of 35 and 55 and verify your answer.

Q22. PQRS is a trapezium in which $PQ \parallel SR$ and its diagonals intersect at O. Show that:

$$\left[\frac{PO}{QO} = \frac{RO}{SO} \right]$$

Q23. A circle is inscribed in $\triangle ABC$ touching AB, BC and AC at P, Q and R respectively. If AB = 12 cm, AR = 8 cm and CR = 6 cm, find BC.

Q24.

(a) Prove that:

$$\left[(\operatorname{cosec} \theta + \sin \theta)(\operatorname{cosec} \theta - \sin \theta) \right. \\ \left. = \cot^2 \theta + \cos^2 \theta \right]$$

OR

(b) Evaluate:

$$\left[\frac{5 \tan^2 30^\circ + 3 \cos^2 45^\circ - 4 \sin^2 30^\circ}{\sqrt{3} \sin 60^\circ \cos 60^\circ + \cot^2 45^\circ} \right]$$

Q25.

(a) A chord of a circle of radius 14 cm subtends a right angle at the centre. Find the area of the minor segment.

OR

(b) The perimeter of a sector of a circle of radius 21 cm is 75 cm. Find the area of the sector.

SECTION C – Short Answer Questions (3 Marks Each)

Q26. Prove that $(\sqrt{5})$ is an irrational number.

Q27.

(a) For what values of m and n does the pair of linear equations

$$\begin{bmatrix} 2x + 3y = 7 \end{bmatrix}$$

$$\begin{bmatrix} m(x + 2y) + n(x - y) = 21 \end{bmatrix}$$

have infinitely many solutions?

OR

(b) The sum of the numerator and denominator of a fraction is 4 more than twice the numerator. If the numerator and denominator are increased by 3, they are in the ratio 2 : 3. Determine the fraction.

Q28. Find the value of p for which

$$\begin{bmatrix} (2p + 1)x^2 - (7p + 2)x + (7p - 3) = 0 \end{bmatrix}$$

has equal roots. Also find these roots.

Q29.

(a) In two concentric circles, a chord of length 24 cm of the larger circle is a tangent to the smaller circle whose radius is 5 cm. Find the radius of the larger circle.

OR

(b) Prove that the intercept of a tangent between two parallel tangents to a circle subtends a right angle at the centre.

Q30. Prove that:

$$\begin{bmatrix} 2(\sin^6 \theta + \cos^6 \theta) \\ \bullet \quad 3(\sin^4 \theta + \cos^4 \theta) \\ \quad + 1 = 0 \end{bmatrix}$$

Q31. If the mean of the following distribution is 54, find the value of p.

Class	0–20	20–40	40–60	60–80	80–100
Frequency	7	p	10	9	13